

BlueRetro: Xbox One gamepad as an NES joystick — oscilloscope verification

Document version: 1.0 · **Date:** 2026-10-01 · **Measurement date:** 2026-09-30

Related document: blueretro-murmulator-project-summary_EN.md, section "Oscilloscope signal verification (confirmed)".

1. Test objective

Verify that BlueRetro (ESP32 D1 mini, firmware v25.04_hw1, NES / GamePad mode) correctly passes Xbox One gamepad button presses to the Murmulator using the NES shift-register protocol: each button appears on the DATA line in its own bit position, in the correct phase relative to CLOCK and LATCH.

2. Oscilloscope connection and settings

Oscilloscope input	Signal	Note
CH1 (yellow)	DATA	In Fig. 1 (01_LATCH_signal) CH1 is connected to LATCH
CH2 (blue)	CLOCK	All figures
EXT TRIG	LATCH	Triggering on the rising edge of LATCH

Parameter	Value
Vertical scale CH1 / CH2	2 V/div / 2 V/div
Timebase	200 μ s/div
Trigger	Ext, rising edge, level 1.08 V
Trigger point position (T marker)	$\approx$ 1 division from the left edge of the screen

Capture order: Fig. 1 — LATCH signal; Fig. 2 — no button pressed; Figs. 3–10 — exactly one button pressed at a time, in NES bit order (from bit 0 to bit 7).

3. Expected NES protocol behavior

- Inverted logic: LOW = button pressed, HIGH = released.
- On the LATCH edge, bit 0 (button A) is placed on DATA immediately — without a clock pulse.
- Each subsequent CLOCK pulse shifts the register: bits 1–7 appear one per clock ($\approx$ 80 μ s per bit).
- Bit order: A, B, Select, Start, Up, Down, Left, Right.

4. Summary of results

Fig.	Xbox One button	NES button	Bit	Result
1	— (LATCH signal)	—	—	Period $\approx$ 1200 μ s, pulse $\approx$ 40 μ s
2	nothing pressed	—	—	DATA = HIGH in all 8 bits ✓
3	A	A	0	LOW at bit position 0 ✓
4	X	B	1	LOW at bit position 1 ✓
5	View	Select	2	LOW at bit position 2 ✓
6	Menu	Start	3	LOW at bit position 3 ✓

Fig.	Xbox One button	NES button	Bit	Result
7	D-pad Up	Up	4	LOW at bit position 4 ✓
8	D-pad Down	Down	5	LOW at bit position 5 ✓
9	D-pad Left	Left	6	LOW at bit position 6 ✓
10	D-pad Right	Right	7	LOW at bit position 7 ✓

Result: all 8 buttons appear on the DATA line in the expected bit position; the Xbox One → NES mapping (A→A, X→B, View→Select, Menu→Start, D-pad→Up/Down/Left/Right) and the bit order are confirmed (confirmed).

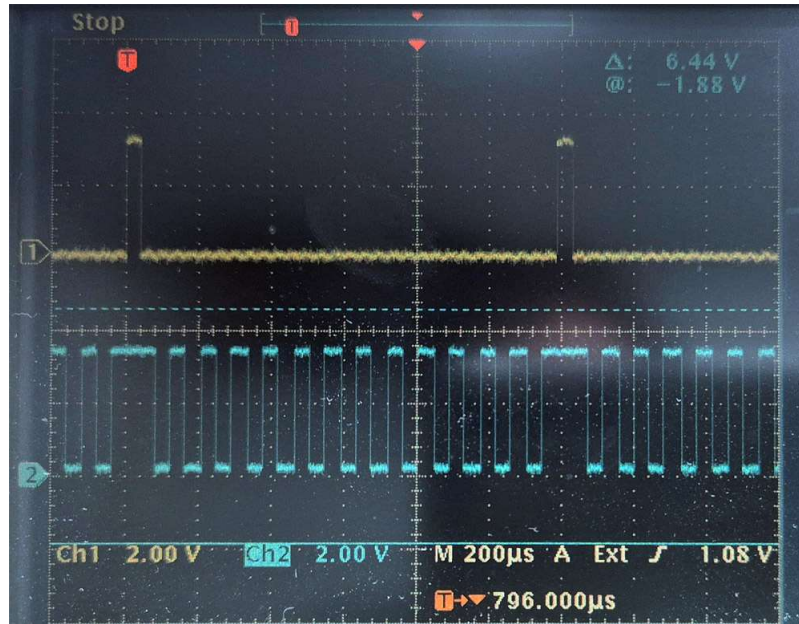
5. Additional observation: CLOCK polling mode

In Figs. 1–2 CLOCK runs continuously and LATCH repeats every $\approx 1200 \mu\text{s}$. In Figs. 3–10 LATCH is followed by a burst of ≈ 12 – 13 CLOCK pulses, after which the CLOCK line stays HIGH and the next LATCH does not appear within the screen window ($\approx 1.8 \text{ ms}$ after the trigger). In other words, the Murmulator polling period differed between captures. This does not affect reading correctness: in both modes there are more than 8 clocks. The reason for the difference has not been established.

6. Oscillograms

Fig. 1. LATCH signal

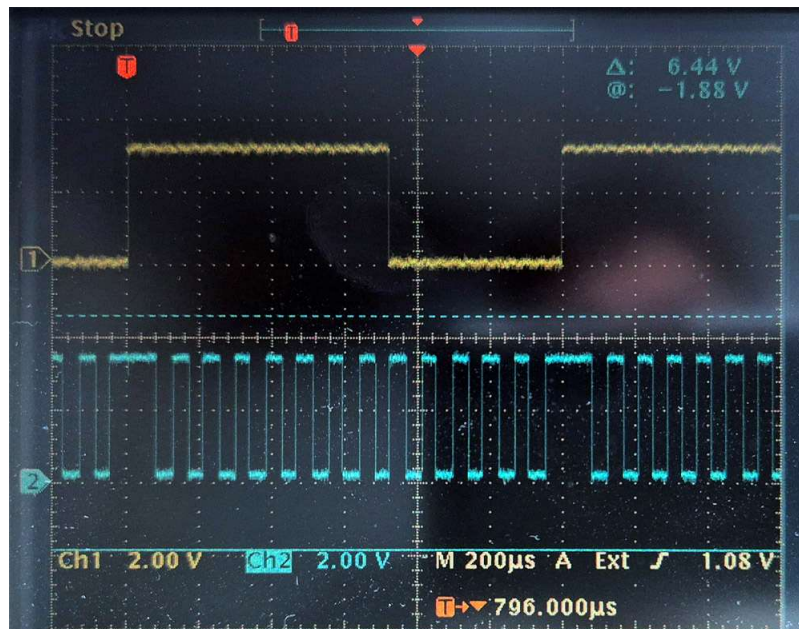
File: 01_LATCH_signal.jpg · CH1 = LATCH, CH2 = CLOCK, EXT TRIG = LATCH



- CH1 (LATCH): short positive pulses $\approx 40 \mu\text{s}$ wide with a period of $\approx 1200 \mu\text{s}$ (≈ 6 divisions) — polling rate $\approx 833 \text{ Hz}$.
- CH2 (CLOCK): a continuous clock train with a period of $\approx 80 \mu\text{s}$ ($\approx 12.5 \text{ kHz}$); around the LATCH pulse the CLOCK line is held HIGH.

Fig. 2. No button pressed

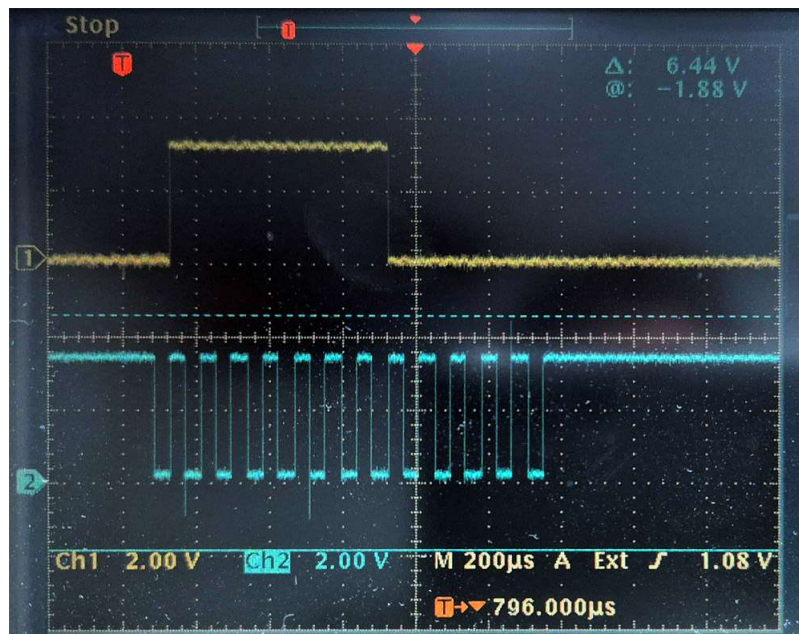
File: 02_no_button_pressed.jpg · CH1 = DATA, CH2 = CLOCK, EXT TRIG = LATCH



- After the LATCH edge the DATA line stays HIGH for all 8 bits ($\approx 700 \mu\text{s}$) — all buttons released.
- After the 8th bit DATA goes LOW and stays there until the next LATCH; at the next LATCH HIGH is loaded again. This matches the original NES controller: after 8 bits the shift register outputs LOW.

Fig. 3. Button A (Xbox A) — bit 0

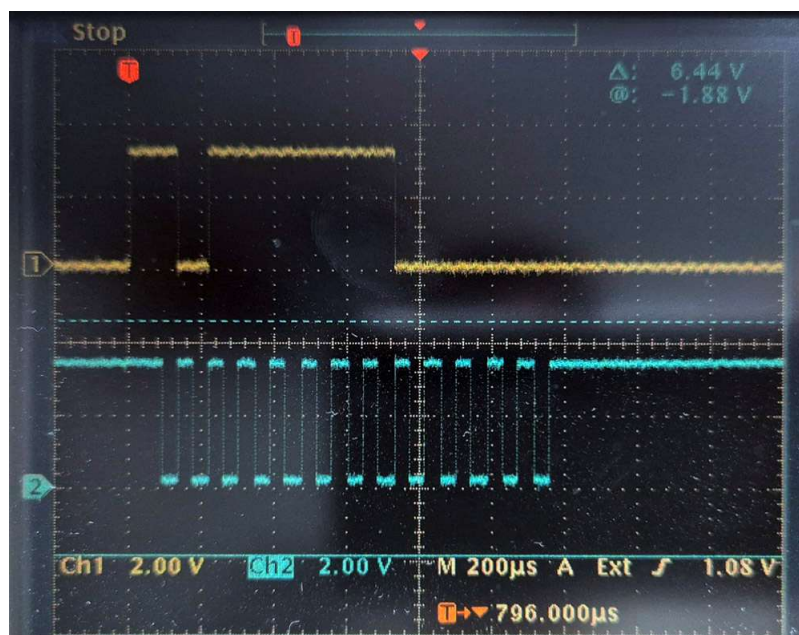
File: 03_A_(xbox_A).jpg · CH1 = DATA, CH2 = CLOCK, EXT TRIG = LATCH



- DATA is LOW immediately after the LATCH edge, before the first CLOCK pulse — bit 0 is placed on the line by the latch, without a clock pulse.
- With the first CLOCK pulse DATA goes HIGH (bits 1–7 — buttons released); after the 8th bit — LOW.

Fig. 4. Button B (Xbox X) — bit 1

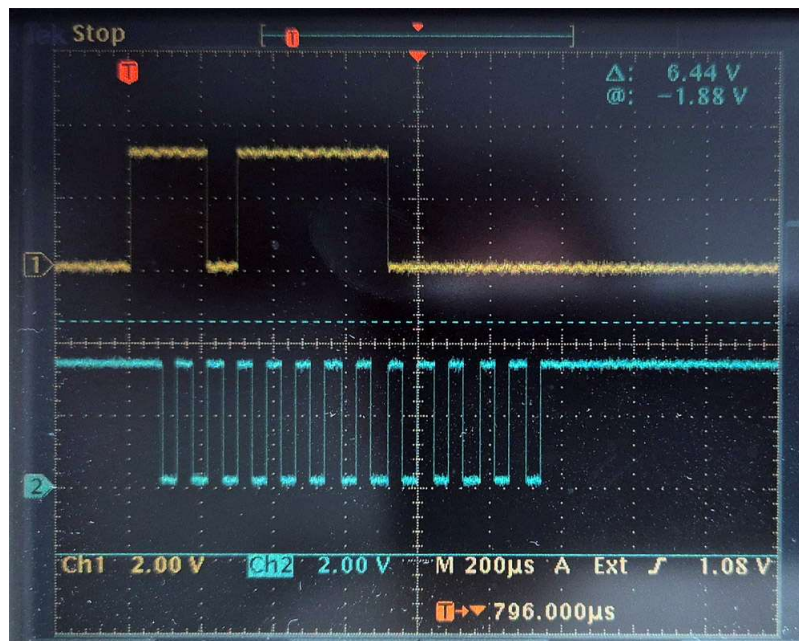
File: 04_B_(xbox_X).jpg · CH1 = DATA, CH2 = CLOCK, EXT TRIG = LATCH



- A LOW pulse $\approx 80 \mu s$ long (one CLOCK period) in the second bit position after LATCH.

Fig. 5. Button Select (Xbox View) — bit 2

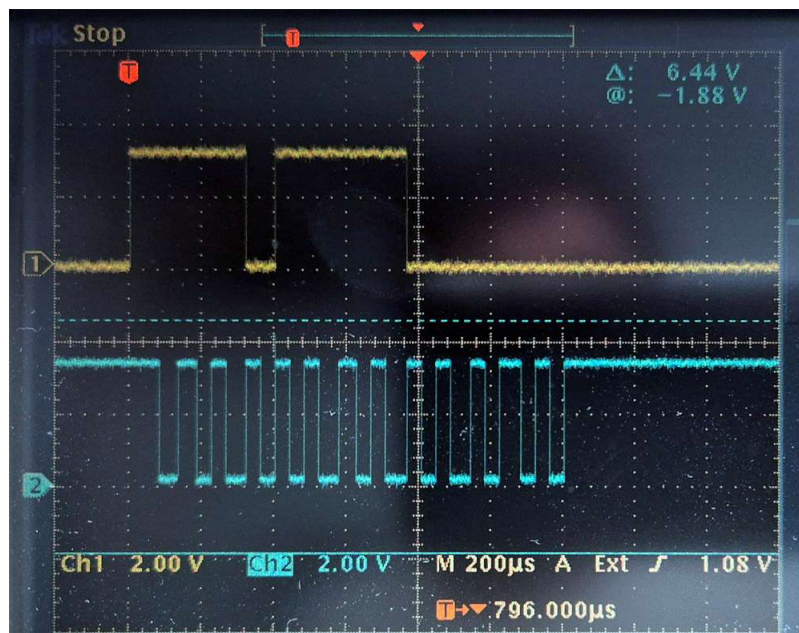
File: 05_SELECT_(xbox_view).jpg · CH1 = DATA, CH2 = CLOCK, EXT TRIG = LATCH



- A $\approx 80 \mu\text{s}$ LOW pulse in the third bit position after LATCH.

Fig. 6. Button Start (Xbox Menu) — bit 3

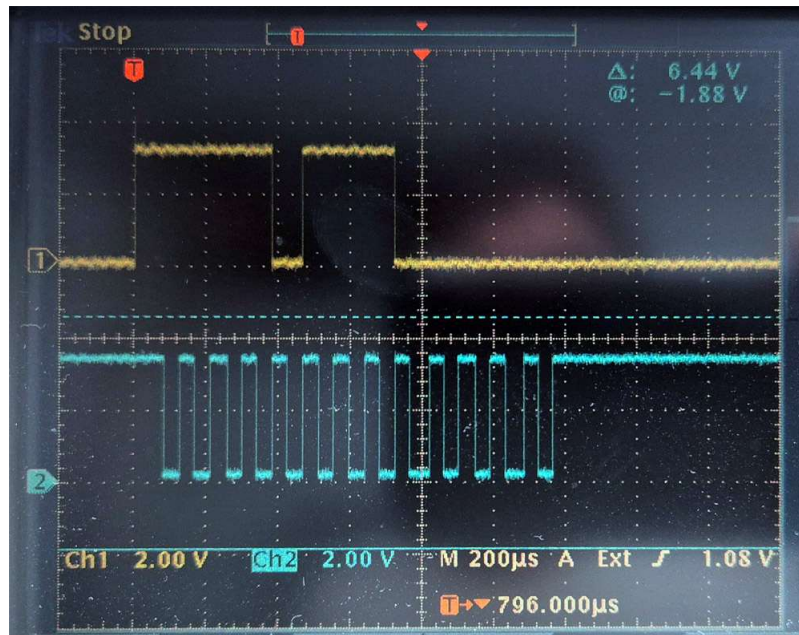
File: 06_START_(xbox_menu).jpg · CH1 = DATA, CH2 = CLOCK, EXT TRIG = LATCH



- A $\approx 80 \mu\text{s}$ LOW pulse in the fourth bit position after LATCH.

Fig. 7. Up (Xbox D-pad Up) — bit 4

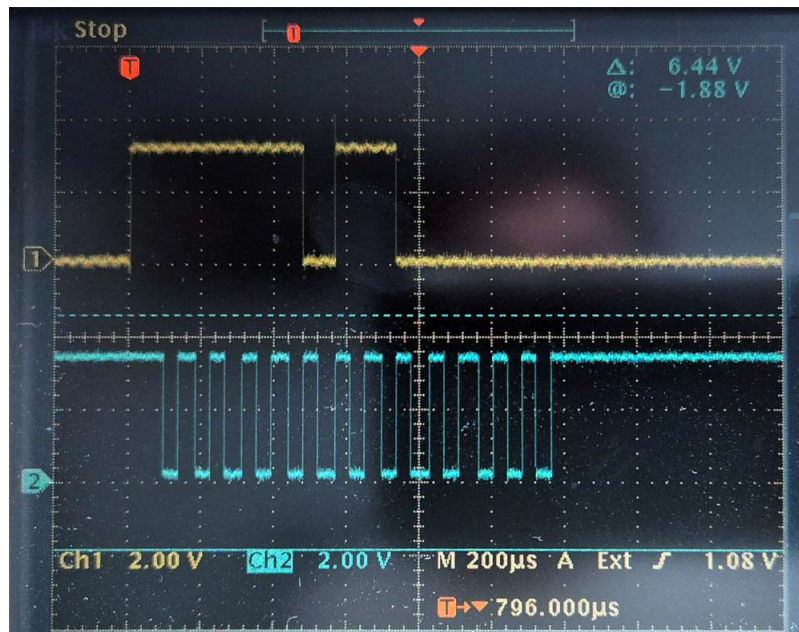
File: 07_up_(xbox_dpad_up).jpg · CH1 = DATA, CH2 = CLOCK, EXT TRIG = LATCH



- A $\approx 80 \mu\text{s}$ LOW pulse in the fifth bit position after LATCH.

Fig. 8. Down (Xbox D-pad Down) — bit 5

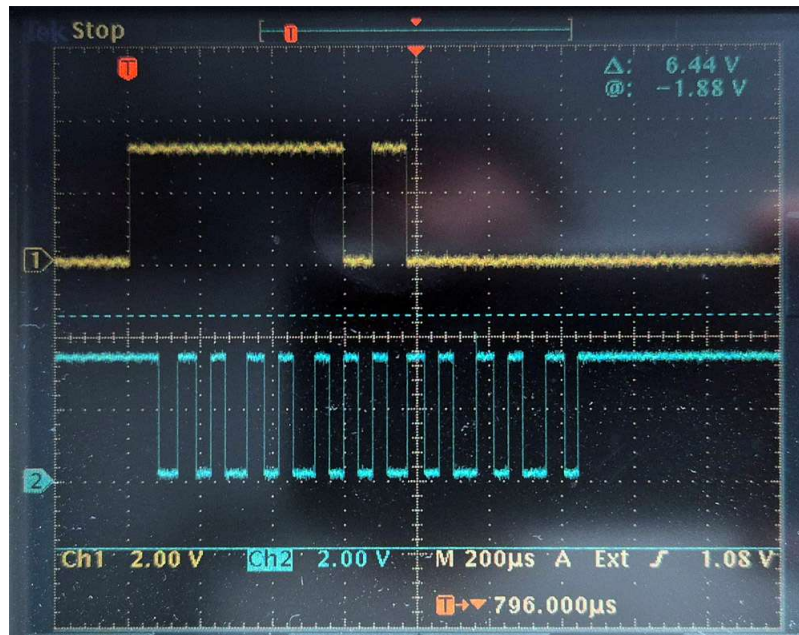
File: 08_down_(xbox_dpad_down).jpg · CH1 = DATA, CH2 = CLOCK, EXT TRIG = LATCH



- A $\approx 80 \mu\text{s}$ LOW pulse in the sixth bit position after LATCH.

Fig. 9. Left (Xbox D-pad Left) — bit 6

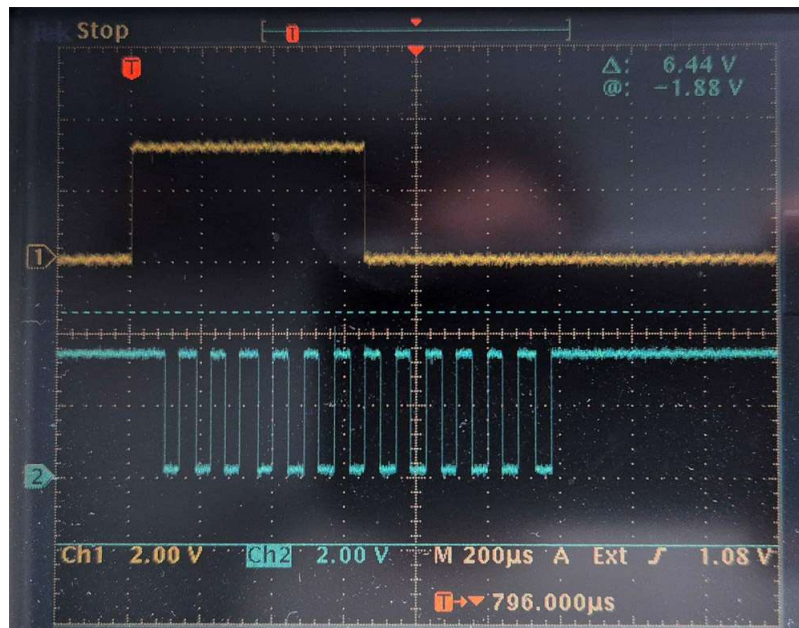
File: 09_left_(xbox_dpad_left).jpg · CH1 = DATA, CH2 = CLOCK, EXT TRIG = LATCH



- A $\approx 80 \mu\text{s}$ LOW pulse in the seventh bit position after LATCH.

Fig. 10. Right (Xbox D-pad Right) — bit 7

File: 10_right_(xbox_dpad_right).jpg · CH1 = DATA, CH2 = CLOCK, EXT TRIG = LATCH



- Bit 7 is the last one, so its LOW merges with the LOW level after the 8th bit: there is no separate pulse; the HIGH section ends one clock earlier than with "no button pressed" (see Fig. 2).